LA HW

Included exercises: 1, 4, 9, 13, 17, 45-46, 51, 53, 60, 62-64, 71, 75

Original Exercise 1

mpute each $\langle f, g \rangle$.

$$f(t) = t^3 \quad \text{and} \quad g(t) = 1$$

Original Exercise 4

$$f(t) = t^2 \quad \text{and} \quad g(t) = t^2$$

Exercises 9-16

In Exercises 9–16, use the Frobenius inner product for $\mathcal{M}_{2\times 2}$ *to compute each* $\langle A, B \rangle$.

Original Exercise 9

npute each $\langle A, B \rangle$.

$$A = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Original Exercise 13

$$A = \begin{bmatrix} -1 & 2 \\ 0 & 4 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 3 & 2 \\ 1 & -1 \end{bmatrix}$$

Exercises 17-24

In Exercises 17–24, use the inner product for $\mathcal{P}_2$ *in Example 5 to compute each* $\langle f(x), g(x) \rangle$.

Original Exercise 17

mpute each $\langle f(x), g(x) \rangle$.

$$f(x) = 3 \quad \text{and} \quad g(x) = -x + 2$$

Original Exercise 45

In Example 1, verify axioms 3 and 4 of the definition of

Original Exercise 46

In Example 2, verify axioms 3 and 4 of the definition of

Exercises 50-57

30. Every orthogonal set in an inner product space is linearly independent.

Original Exercise 51

$V = \mathcal{R}^n$ and $\langle \mathbf{u}, \mathbf{v} \rangle = 2(\mathbf{u} \cdot \mathbf{v})$

Original Exercise 53

Let $V = \mathrm{C}([0, 2])$, and

$$\langle f, g \rangle = \int_0^1 f(t)g(t)\,dt$$

for all f and g in V. (Note that the limits of integration

Original Exercise 60

$$\cos a \cos b = \frac{\quad}{2}.$$

(a) Use the methods of Example 5 to obtain $p_3(x)$, the normalized Legendre polynomial of degree 3.

(b) Use the result of (a) to find the least-squares approximation of $\sqrt[3]{x}$ on $[-1, 1]$ as a polynomial of degree

Original Exercise 62

Suppose that $\langle \mathbf{u}, \mathbf{v} \rangle$ is an inner product for a vector space V. For any scalar $r > 0$, define $\langle\langle \mathbf{u}, \mathbf{v} \rangle\rangle = r \langle \mathbf{u}, \mathbf{v} \rangle$.

(a) Prove that $\langle\langle \mathbf{u}, \mathbf{v} \rangle\rangle$ is an inner product on V.

(b) Why is $\langle\langle \mathbf{u}, \mathbf{v} \rangle\rangle$ not an inner product if $r \leq 0$?

ercises 63–70, let **u**, **v**, *and* **w** *be vectors in an inner product*

Original Exercise 63

V, and let c be a scalar.

Prove that $\|\mathbf{v}\| = 0$ if and only if $\mathbf{v} = \mathbf{0}$.

Original Exercise 64

Prove that $\|c\mathbf{v}\| = |c|\|\mathbf{v}\|$.

Exercise 71

(b) Use (a) and Exercise 71 to prove the converse of Exercise 49: For any inner product on $\mathcal{R}^n$, there exists a positive definite matrix A such that

Original Exercise 71

Let V be a finite-dimensional inner product space, and suppose that $\mathcal{B}$ is an orthonormal basis for V. Prove that, for any vectors **u** and **v** in V,

$$\langle \mathbf{u}, \mathbf{v} \rangle = [\mathbf{u}]_{\mathcal{B}} \cdot [\mathbf{v}]_{\mathcal{B}}.$$

Original Exercise 75

Consider the inner product space $\mathcal{M}_{2\times 2}$ with the Frobenius inner product.

(a) Find an orthonormal basis for the subspace of 2×2 symmetric matrices.

(b) Use (a) to find the 2×2 symmetric matrix that is closest to

$$\begin{bmatrix} 1 & 2 \\ 4 & 8 \end{bmatrix}.$$